

①. $A_0, \dots, A_m$ vertices of a polygon. Det. $\vec{A_0A_1} + \vec{A_1A_2} + \dots + \vec{A_{m-1}A_m} + \vec{A_mA_0}$

$$\vec{A_0A_1} + \vec{A_1A_2} = \vec{A_0A_2}$$

$$\vec{A_0A_2} + \vec{A_2A_3} = \vec{A_0A_3}$$

$\vdots$

$$\vec{A_0A_m} + \vec{A_mA_0} = \vec{0} = \vec{A_0A_0}$$

② Can u, v, w be represented with the vertices of a Δ ?

a) $u(7,3), v(-2,-8), w(-5,5)$

$$\boxed{\vec{u} + \vec{v} + \vec{w} = \vec{0} \Rightarrow \text{they form a } \Delta}$$

$$(7,3) + (-2,-8) + (-5,5) = (0,0) \Rightarrow \exists \Delta$$

✓ b) $u(7,3), v(2,8), w(-5,5)$

We observe that $\vec{u} + \vec{w} = \vec{v} \Leftrightarrow \vec{u} + (-\vec{v}) + \vec{w} = \vec{0} \Rightarrow \exists \Delta$.

c) $\|u\| = 7, \|v\| = 3, \|w\| = 11$.

$$\|u\| + \|v\| = 7 + 3 = 10$$

$$\|w\| = 11$$

Δ inequality: $\|u\| + \|v\| \not\geq \|w\|$

$$\Leftrightarrow 10 \geq 11 \begin{matrix} \text{true} \\ \text{false} \end{matrix} \Rightarrow \nexists \Delta$$

d) $u(1,0,1), v(0,1,0), w(2,2,2)$

$$u+v = (1,1,1) \neq (2,2,2) \\ \neq (-2,-2,-2)$$

$$u-v = (1,-1,1) \neq (2,2,2) \neq (-2,-2,-2)$$

$$-u+v = (-1,1,-1) \neq (2,2,2) \neq (-2,-2,-2)$$

$$-u-v = (-1,-1,-1) \neq (2,2,2) \neq (-2,-2,-2)$$

$$\Rightarrow \nexists \Delta$$

✓ e) $u(1,2,-1), v(2,-1,0), w(1,-3,1)$

$$\vec{u} + \vec{w} = \vec{v}$$

OR $\begin{vmatrix} 1 & 2 & -1 \\ 2 & -1 & 0 \\ 1 & -3 & 1 \end{vmatrix} \xrightarrow{L_3+L_1 \rightarrow L_1} \begin{vmatrix} 2 & -1 & 0 \\ 2 & -1 & 0 \\ 1 & -3 & 1 \end{vmatrix} = 0 \Rightarrow \text{lin. dependent} \Rightarrow \exists \Delta$

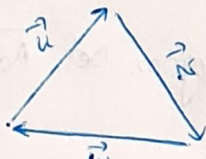
a) another method:

$$u(7,3), v(2,8), w(-5,5)$$

$$\vec{u} = 7\vec{i} + 3\vec{j}$$

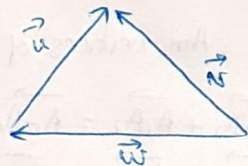
$$\vec{v} = 2\vec{i} + 8\vec{j}$$

$$\vec{w} = -5\vec{i} + 5\vec{j}$$



$$\vec{u} + \vec{v} = -\vec{w}$$

$$\vec{u} + \vec{v} + \vec{w} = 0$$



$$\vec{u} + \vec{w} = -\vec{v}$$

$$\vec{u} - \vec{v} + \vec{w} = 0$$

$$\vec{u} - \vec{v} + \vec{w} = 7\vec{i} + 3\vec{j} - 2\vec{i} - 8\vec{j} - 5\vec{i} + 5\vec{j} = \vec{0}$$

$\Rightarrow \vec{u}, \vec{v}, \vec{w}$ can be represented as vertices of a triangle.

e b) another method:

$$u(1,2,-1), v(2,-1,0), w(1,-3,1)$$

$$\|u\| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}$$

$$\|v\| = \sqrt{2^2 + (-1)^2 + 0^2} = \sqrt{5}$$

$$\|u+v\| = \sqrt{(1+2)^2 + (2-1)^2 + (-1+0)^2} = \sqrt{3^2 + 1^2 + (-1)^2} = \sqrt{11} \leq \|u\| + \|v\| = \sqrt{6} + \sqrt{5}$$

$$\vec{u} - \vec{v} + \vec{w} = \vec{0}$$

$$\vec{u} + 2\vec{v} - \vec{w} - 2\vec{i} + \vec{j} + \vec{i} - 3\vec{j} + \vec{k} = \vec{0}$$

$\Rightarrow \vec{u} - \vec{v} + \vec{w}$ can be represented as a triangle.

③ ABCDEF regular hexagon centered at O.

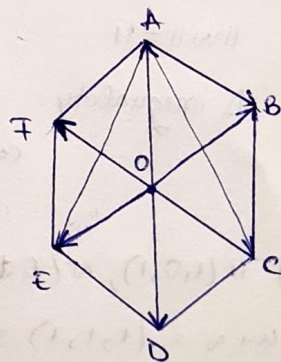
a) $\vec{OA}, \vec{OB}, \vec{OC}, \vec{OD}$ in terms of $\vec{OE}$ and $\vec{OF}$.

$$\vec{OB} = -\vec{OE}$$

$$\vec{OC} = -\vec{OF}$$

$$\vec{OE} = \vec{OF} + \vec{OD} \Rightarrow \vec{OD} = \vec{OE} - \vec{OF}$$

$$\vec{OA} = \vec{OF} + \vec{OB} = \vec{OF} - \vec{OE}$$



b) Show: $\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} = 3\vec{AD}$ (1)

$$\vec{AD} = \vec{OD} - \vec{OA}$$

$$\vec{AD} = \vec{AO} + \vec{OD}$$

$$\vec{AD} = \vec{OB} - \vec{OA}$$

$$(1) \Rightarrow \vec{OB} - \vec{OA} + \vec{OC} - \vec{OA} + \vec{OD} - \vec{OA} + \vec{OE} - \vec{OA} + \vec{OF} - \vec{OA} = 3(\vec{OB} - \vec{OA})$$

$$\Rightarrow \vec{OB} + \vec{OC} + \vec{OD} + \vec{OE} + \vec{OF} - 5\vec{OA} = 3(\vec{OB} - \vec{OA})$$

$$\Rightarrow -\vec{OA} - \vec{OF} + \vec{OB} + \vec{OE} + \vec{OF} - 5\vec{OA} = 3(\vec{OB} - \vec{OA})$$

$$\Rightarrow -\vec{OA} - 5\vec{OA} = 3(-\vec{OA} - \vec{OA})$$

$$\Rightarrow -6\vec{OA} = -6\vec{OA} \text{ true.}$$

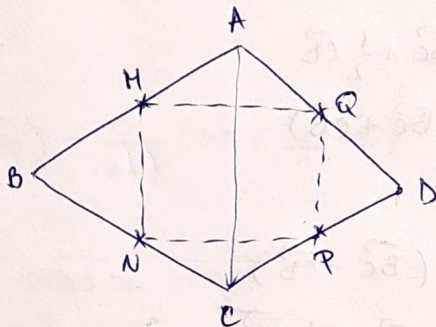
④ ABCD quadrilateral.

M, N, P, Q midpoints of [AB], [BC], [CD], [DA].

$$\vec{MN} + \vec{PQ} = \vec{0} \quad ?$$

* midpoints of sides of arbitrary quadrilateral $\rightarrow$ parallelogram.

$$\vec{MN} + \vec{PQ} = \vec{0}$$



$$\vec{MN} + \vec{PQ} = \vec{0}$$

$$\Leftrightarrow \vec{MB} + \vec{BN} + \vec{PD} + \vec{DQ} = \vec{0}$$

$$\Leftrightarrow \frac{1}{2} \vec{AB} + \frac{1}{2} \vec{BC} + \frac{1}{2} \vec{CD} + \frac{1}{2} \vec{DA} = \vec{0}$$

$$\Leftrightarrow \frac{1}{2} (\vec{AB} + \vec{BC}) + \frac{1}{2} (\vec{CD} + \vec{DA}) = \vec{0}$$

$$\Leftrightarrow \frac{1}{2} \vec{AC} + \frac{1}{2} \vec{CA} = \vec{0} \quad | \cdot 2$$

$$\Leftrightarrow \vec{AC} = -\vec{CA} = \vec{AC} \quad \text{true}$$

$$\vec{MN} + \vec{PQ} = \vec{0}$$

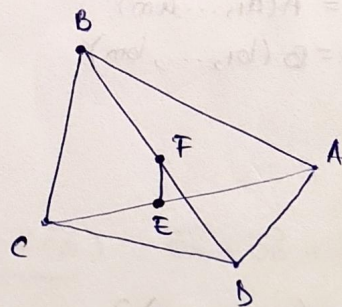
$$\Rightarrow \vec{MN} = \vec{QP} \quad \left. \begin{array}{l} M \neq Q \\ N \neq P \end{array} \right\} \Rightarrow \text{MNPQ parallelogram.}$$

⑤ ABCD quadrilateral

E - mid. [AC]

F - mid. [BD]

$$\text{Show : } \vec{EF} = \frac{1}{2} (\vec{AB} + \vec{CD}) = \frac{1}{2} (\vec{AD} + \vec{CB})$$



$$\vec{EF} = \frac{1}{2} (\vec{AB} + \vec{CD}) \quad ?$$

$$\vec{EF} = \vec{EC} + \vec{CB} + \vec{BF} = \frac{1}{2} \vec{AC} + \vec{CB} + \frac{1}{2} \vec{BD} =$$

$$= \frac{1}{2} \vec{AC} + \frac{1}{2} \vec{CB} + \frac{1}{2} \vec{CB} + \frac{1}{2} \vec{BD} = \frac{1}{2} (\vec{AC} + \vec{CB}) + \frac{1}{2} \vec{CB} + \frac{1}{2} \vec{BD} =$$

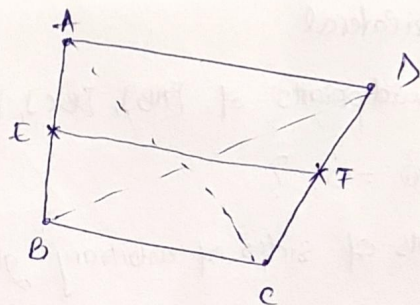
$$= \frac{1}{2} \vec{AB} + \frac{1}{2} \vec{CB} + \frac{1}{2} \vec{BD} = \frac{1}{2} \vec{AB} + \frac{1}{2} (\vec{CB} + \vec{BD}) = \frac{1}{2} \vec{AB} + \frac{1}{2} \vec{CD}$$

⑥ ABCD quadrilateral.

E - mid. [AB]

F - mid. [CD]

Show: $\vec{EF} = \frac{1}{2}(\vec{AD} + \vec{BC})$



$$\vec{EF} = \vec{EB} + \vec{BC} + \vec{CF}$$

$$= \frac{1}{2}\vec{AB} + \vec{BC} + \frac{1}{2}\vec{CD}$$

$$= \frac{1}{2}\vec{AB} + \frac{1}{2}\vec{BC} + \frac{1}{2}\vec{BC} + \frac{1}{2}\vec{CD}$$

$$= \frac{1}{2}(\vec{AB} + \vec{BC}) + \frac{1}{2}(\vec{BC} + \vec{CD})$$

$$= \frac{1}{2}\vec{AC} + \frac{1}{2}\vec{BD}$$

$$= \frac{1}{2}(\vec{AB} + \vec{DC}) + \frac{1}{2}(\vec{BC} + \vec{CB})$$

$$= \frac{1}{2}\vec{AB} + \frac{1}{2}\vec{DC} + \frac{1}{2}\vec{BC} + \frac{1}{2}\vec{CB}$$

$$= \frac{1}{2}\vec{AD} + \frac{1}{2}\vec{BC}$$

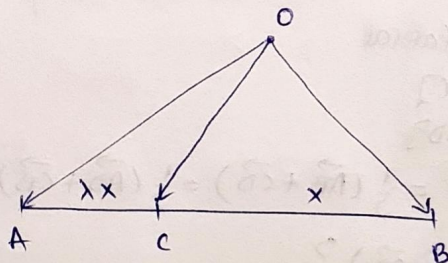
⑦ $\lambda = |CA| : |CB|, C \in [AB]$

Show: $\vec{OC} = \frac{1}{1+\lambda}\vec{OA} + \frac{\lambda}{1+\lambda}\vec{OB}$

Deduce C coordinates: $\left(\frac{a_1 + \lambda b_1}{1+\lambda}, \frac{a_2 + \lambda b_2}{1+\lambda}, \dots, \frac{a_m + \lambda b_m}{1+\lambda}\right)$

$A = A(a_1, \dots, a_m)$

$B = B(b_1, \dots, b_m)$



$$\left. \begin{array}{l} A-C-B \text{ (collinear)} \\ \lambda = |CA| : |CB| \end{array} \right\} \Rightarrow \vec{AC} = \lambda \cdot \vec{CB} \Leftrightarrow \vec{OC} - \vec{OA} = \lambda(\vec{OB} - \vec{OC})$$

$$\Leftrightarrow \vec{OC} + \lambda \vec{OC} = \vec{OA} + \lambda \vec{OB}$$

$$\Leftrightarrow (1+\lambda)\vec{OC} = \vec{OA} + \lambda \vec{OB} \quad | : (\lambda+1)$$

$$\Leftrightarrow \vec{OC} = \frac{1}{\lambda+1}\vec{OA} + \frac{\lambda}{\lambda+1}\vec{OB}$$

$$\Rightarrow C = \frac{1}{\lambda+1}(A + \lambda B)$$

$$\Rightarrow C = \frac{A + \lambda B}{\lambda+1} \Rightarrow$$

$$\left\{ \begin{array}{l} c_1 = \frac{a_1 + \lambda b_1}{\lambda+1} \\ c_2 = \frac{a_2 + \lambda b_2}{\lambda+1} \\ \dots \\ c_m = \frac{a_m + \lambda b_m}{\lambda+1} \end{array} \right.$$

- 5) ABC triangle
 $D \in [BC]$ s.t. AD bisector.

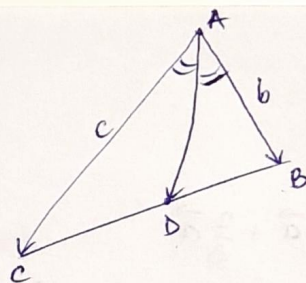
Express $\vec{AD}$ using $b = \vec{AB}$ and $c = \vec{AC}$.

Bisector Theorem: $\frac{AC}{AB} = \frac{CD}{DB}$

$$\Rightarrow \frac{c}{b} = \frac{CD}{DB} = k$$

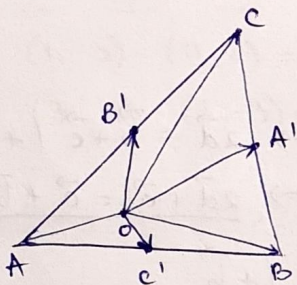
$$\vec{AD} = \frac{1}{k+1} (\vec{AC} + k \vec{AB})$$

$$\vec{AD} = \frac{1}{\frac{b}{c} + 1} (\vec{AC} + \frac{b}{c} \vec{AB}) = \frac{c}{b+c} (\vec{AC} + \frac{b}{c} \vec{AB})$$



- 9) A', B', C' midpoints of sides of $\triangle ABC$.

Show: $\vec{OA'} + \vec{OB'} + \vec{OC'} = \vec{OA} + \vec{OB} + \vec{OC}$, (*) O.



$$\vec{OA'} = \vec{OB} + \vec{OC} = \vec{OB} + \frac{1}{2} \vec{BC}$$

$$\vec{OB'} = \vec{OC} + \vec{OA} = \vec{OC} + \frac{1}{2} \vec{CA}$$

$$\vec{OC'} = \vec{OA} + \vec{OB} = \vec{OA} + \frac{1}{2} \vec{AB}$$

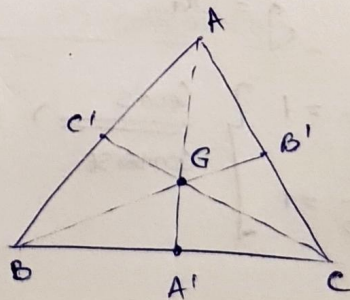
$$\vec{OA'} + \vec{OB'} + \vec{OC'} = \vec{OA} + \vec{OB} + \vec{OC} + \frac{1}{2} (\vec{AB} + \vec{BC} + \vec{CA}) = \vec{OA} + \vec{OB} + \vec{OC}.$$

- 10) Show that medians in a Δ intersect in one point and deduce the ratio in which the common intersection point divides the medians.

$$\begin{aligned} \vec{AG} &= \vec{AC} + \vec{CA'} + \vec{A'G} = \vec{AC} + \frac{1}{2} \vec{CB} + \vec{A'G} = \\ &= \frac{1}{2} \vec{AC} + \frac{1}{2} (\vec{AC} + \vec{CB}) + \vec{A'G} \\ &= \frac{1}{2} (\vec{AC} + \vec{BA}) + \vec{A'G} \end{aligned}$$

$$\vec{BB'} = \frac{1}{2} (\vec{BC} + \vec{BA})$$

$$\vec{CC'} = \frac{1}{2} (\vec{CA} + \vec{CB})$$



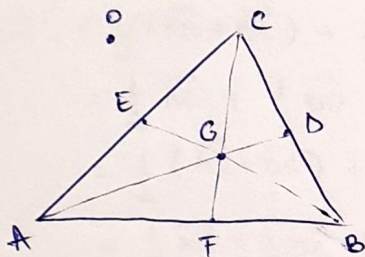
$$\vec{OA} = \vec{a}$$

$$\vec{OB} = \vec{b}$$

$$\vec{OG} = \frac{1}{3}\vec{a} + \frac{2}{3}\vec{a}$$

$$\left(\frac{1}{3} + \frac{2}{3}\right)\vec{OG} = \frac{1}{3}\vec{a} + \frac{2}{3}\vec{a}$$

$$\Rightarrow \vec{AG} = 2\vec{A'G}$$



D - mid [BC]

F - mid [AB]

E - mid [AC]

• the medians are concurrent

$$\vec{OA} = \vec{a} \quad \vec{OC} = \vec{c} \quad \vec{OE} = \vec{e}$$

$$\vec{OB} = \vec{b} \quad \vec{OD} = \vec{d} \quad \vec{OF} = \vec{f}$$

$$\vec{OD} = \vec{d} = \frac{\vec{OC} + \vec{OB}}{2} = \frac{\vec{c} + \vec{b}}{2} \Rightarrow 2\vec{d} = \vec{b} + \vec{c} \quad | + \vec{a}$$

$$\Rightarrow \underline{2\vec{d} + \vec{a} = \vec{a} + \vec{b} + \vec{c}}$$

$$\vec{e} = \frac{\vec{c} + \vec{a}}{2} \Rightarrow 2\vec{e} = \vec{a} + \vec{c} \quad | + \vec{b}$$

$$\Rightarrow \underline{2\vec{e} + \vec{b} = \vec{a} + \vec{b} + \vec{c}}$$

$$\vec{f} = \frac{\vec{a} + \vec{b}}{2} \Rightarrow 2\vec{f} = \vec{a} + \vec{b} \quad | + \vec{c}$$

$$\Rightarrow \underline{2\vec{f} + \vec{c} = \vec{a} + \vec{b} + \vec{c}}$$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} = 2\vec{d} + \vec{a} = 2\vec{e} + \vec{b} = 2\vec{f} + \vec{c} \quad | \cdot \frac{1}{3}$$

$$\Rightarrow \frac{\vec{a} + \vec{b} + \vec{c}}{3} = \frac{2\vec{d} + \vec{a}}{3} = \frac{2\vec{e} + \vec{b}}{3} = \frac{2\vec{f} + \vec{c}}{3}$$

$$\vec{g} = \vec{OG}$$

$$\left. \begin{array}{l} \frac{CE}{EA} = 1 \\ \frac{AF}{FB} = 1 \end{array} \right\} \text{Ceva's Converse} \Rightarrow CF \cap AD \cap BE = \{G\}$$

ii) Collinear points?

a) $P(3, -5), Q(-1, 2), R(-5, 9)$

$$\begin{aligned} m_{PQ} &= \frac{2 - (-5)}{-1 - 3} = \frac{7}{-4} \\ m_{QR} &= \frac{9 - 2}{-5 - (-1)} = \frac{7}{-4} \end{aligned} \quad \Rightarrow P, Q, R \text{ collinear.}$$

$$\begin{aligned} \vec{AB} &= \vec{r}_B - \vec{r}_A = (-1, 2) - (3, -5) = (-4, 7) \\ \vec{BC} &= \vec{r}_C - \vec{r}_B = (-5, 9) - (-1, 2) = (-4, 7) \end{aligned} \quad \Rightarrow \vec{AB} = 1 \cdot \vec{BC} \Rightarrow \text{collinear } A, B, C. \checkmark$$

b) $A(11, 2), B(1, -3), C(31, 13)$

$$\begin{aligned} m_{AB} &= \frac{-3 - 2}{1 - 11} = \frac{-5}{-10} = \frac{1}{2} \\ m_{BC} &= \frac{13 - (-3)}{31 - 1} = \frac{16}{30} = \frac{8}{15} \end{aligned} \quad \Rightarrow A, B, C \text{ not collinear.}$$

$$\begin{aligned} \vec{AB} &= \vec{r}_B - \vec{r}_A = (1, -3) - (11, 2) = (-10, -5) \\ \vec{BC} &= \vec{r}_C - \vec{r}_B = (31, 13) - (1, -3) = (30, 16) \end{aligned} \quad \Rightarrow \text{not collinear.}$$

c) $P(1, 0, -1), Q(0, -1, 2), R(-1, -2, 5)$

$d_{PQ} =$

$d_{QP} =$

$d_{PR} =$

$$\begin{aligned} \vec{PQ} &= (0, -1, 2) - (1, 0, -1) = (-1, -1, 3) \\ \vec{QR} &= (-1, -2, 5) - (0, -1, 2) = (-1, -1, 3) \end{aligned} \quad \Rightarrow \text{lin. dependent} \Rightarrow P, Q, R \text{ collinear.}$$

d) $A(-1, -1, 4), B(1, 1, 0), C(2, 2, 2)$

$$\begin{aligned} \vec{AB} &= (1, 1, 0) - (-1, -1, 4) = (2, 2, -4) \\ \vec{BC} &= (2, 2, 2) - (1, 1, 0) = (1, 1, 2) \end{aligned} \quad \Rightarrow \text{lin. dependent} \Rightarrow \text{collinear.}$$

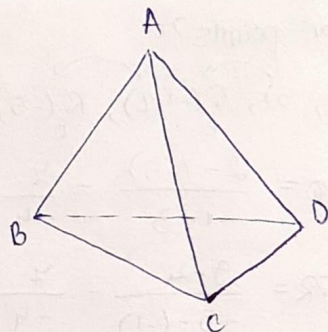
(12) ABCD tetrahedron.

a) $\vec{AB} + \vec{BC} + \vec{CB} = ?$

$$\vec{AB} + \vec{BC} + \vec{CB} = \vec{AB} + \vec{CB} = \vec{AC}$$

b) $\vec{AB} + \vec{BC} + \vec{DB} = \vec{AB} + \vec{DC} = \vec{AC}$

c) $\vec{AB} + \vec{CB} + \vec{BC} + \vec{DA} = \vec{AC} + \vec{CA} = \vec{0}$



(13) ABCD tetrahedron.

Show: $\vec{AB} + \vec{BC} = \vec{BD} + \vec{AC}$

$$\Rightarrow \vec{AB} - \vec{BD} + \vec{BC} - \vec{AC} = \vec{0}$$

$$\Rightarrow \vec{AB} + \vec{DB} + \vec{BC} + \vec{CA} = \vec{0}$$

$$\Rightarrow \vec{AB} + \vec{BA} = \vec{0}$$

$$\Rightarrow \vec{0} = \vec{0} \text{ true.}$$

(14) SABCD pyramid with apex S and base ABCD (par.)

Show: $\vec{SA} + \vec{SB} + \vec{SC} + \vec{SD} = 4\vec{SO}$; O - center of par.

$$\vec{SO} = \vec{SA} + \vec{AO}$$

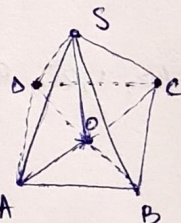
$$\vec{SO} = \vec{SB} + \vec{BO}$$

$$\vec{SO} = \vec{SC} + \vec{CO}$$

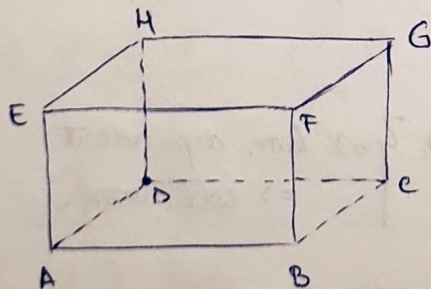
$$\vec{SO} = \vec{SD} + \vec{DO}$$

$$\text{(+)} \quad 4\vec{SO} = \vec{SA} + \vec{SB} + \vec{SC} + \vec{SD} + \vec{AO} + \vec{CO} + \vec{BO} + \vec{DO}$$

$$4\vec{SO} = \vec{SA} + \vec{SB} + \vec{SC} + \vec{SD} + \underbrace{\vec{AO} + \vec{CO}}_{\vec{0}} + \underbrace{\vec{BO} + \vec{DO}}_{\vec{0}} = \vec{SA} + \vec{SB} + \vec{SC} + \vec{SD}$$



(15) Coordinates of the vertices of the parallelepiped whose faces lie in the coord. planes and in the planes $x=1, y=3, z=-2$.



D(0,0,0)

A(1,0,0)

B(1,3,0)

C(0,3,0)

H(0,0,-2)

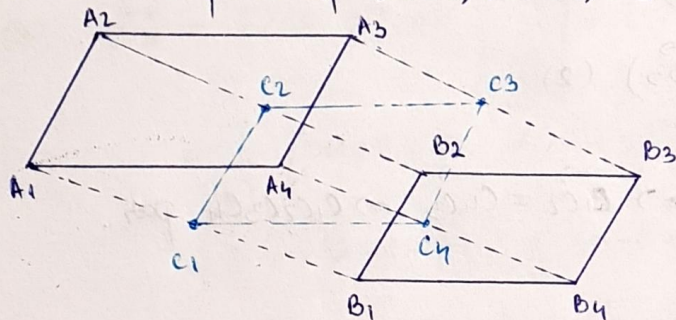
E(1,0,-2)

F(1,3,-2)

G(0,3,-2)

⑤ E^3 ; $A_1 A_2 A_3 A_4$ } parallelograms $\Rightarrow \begin{cases} \overrightarrow{A_1 A_2} = \overrightarrow{A_4 A_3} ; \overrightarrow{A_1 A_4} = \overrightarrow{A_2 A_3} \\ \overrightarrow{B_1 B_2} = \overrightarrow{B_4 B_3} ; \overrightarrow{B_1 B_4} = \overrightarrow{B_2 B_3} \end{cases}$
(1)

Show: midpoints of $[A_1 B_1], [A_2 B_2], [A_3 B_3], [A_4 B_4]$ vertices of a par.



C_1, C_2, C_3, C_4 midpoints.

(H1). $C_1 C_2 C_3 C_4$ par. $\Rightarrow \begin{cases} \overrightarrow{C_1 C_2} = \overrightarrow{C_4 C_3} \\ \overrightarrow{C_4 C_1} = \overrightarrow{C_3 C_2} \end{cases}$

(*) $\overrightarrow{C_1 C_2} = \overrightarrow{C_4 C_3} \Leftrightarrow \overrightarrow{OC_2} - \overrightarrow{OC_1} = \overrightarrow{OC_3} - \overrightarrow{OC_4}$, O - arbitrary fixed point.

$$\overrightarrow{OC_2} = \frac{\overrightarrow{OA_2} + \overrightarrow{OB_2}}{2}$$

$$\overrightarrow{OC_1} = \frac{\overrightarrow{OA_1} + \overrightarrow{OB_1}}{2}$$

$$\overrightarrow{OC_3} = \frac{\overrightarrow{OA_3} + \overrightarrow{OB_3}}{2}$$

$$\overrightarrow{OC_4} = \frac{\overrightarrow{OA_4} + \overrightarrow{OB_4}}{2}$$

(*) $\Leftrightarrow \frac{\overrightarrow{OA_2} + \overrightarrow{OB_2} - \overrightarrow{OA_1} - \overrightarrow{OB_1}}{2} = \frac{\overrightarrow{OA_3} + \overrightarrow{OB_3} - \overrightarrow{OA_4} - \overrightarrow{OB_4}}{2} \quad | \cdot 2$

$$\Leftrightarrow \overrightarrow{OA_2} - \overrightarrow{OA_1} + \overrightarrow{OB_2} - \overrightarrow{OB_1} = \overrightarrow{OA_3} - \overrightarrow{OA_4} + \overrightarrow{OB_3} - \overrightarrow{OB_4}$$

$$\Leftrightarrow \overrightarrow{A_1 A_2} + \overrightarrow{B_1 B_2} = \overrightarrow{A_4 A_3} + \overrightarrow{B_4 B_3} \text{ true (1)}$$

$\Rightarrow C_1 C_2 C_3 C_4$ par.

(H2) $\overrightarrow{A_1 A_2} + \overrightarrow{A_2 C_2} + \overrightarrow{C_2 C_1} + \overrightarrow{C_1 A_1} = 0$

$$\overrightarrow{B_1 B_2} + \overrightarrow{B_2 C_2} + \overrightarrow{C_2 C_1} + \overrightarrow{C_1 B_1} = 0$$

$$\frac{1}{2} (\overrightarrow{A_1 A_2} + \overrightarrow{B_1 B_2}) = \overrightarrow{C_1 C_2} \quad (+)$$

$$\Rightarrow \frac{1}{2} (\overrightarrow{A_1 A_2} + \overrightarrow{B_1 B_2}) = \overrightarrow{C_1 C_2} \quad (1)$$

$$\overrightarrow{A_4 A_3} + \overrightarrow{A_3 C_3} + \overrightarrow{C_3 C_4} + \overrightarrow{C_4 A_4} = 0.$$

$$\overrightarrow{B_4 B_3} + \overrightarrow{B_3 C_3} + \overrightarrow{C_3 C_4} + \overrightarrow{C_4 B_4} = 0$$

$$\hline \Rightarrow 2 \overrightarrow{C_3 C_4} + \overrightarrow{A_4 A_3} + \overrightarrow{B_4 B_3} = 0 \quad (*)$$

$$\Rightarrow 2 \overrightarrow{C_3 C_4} + \overrightarrow{A_4 A_3} + \overrightarrow{B_4 B_3} = 0$$

$$\Rightarrow \overrightarrow{C_4 C_3} = \frac{1}{2} (\overrightarrow{A_4 A_3} + \overrightarrow{B_4 B_3}) \quad (2)$$

$$(1), (2) \Rightarrow \begin{cases} \overrightarrow{A_1 A_2} = \overrightarrow{A_4 A_3} \\ \overrightarrow{B_1 B_2} = \overrightarrow{B_4 B_3} \end{cases} \Rightarrow \overrightarrow{C_1 C_2} = \overrightarrow{C_4 C_3} \Rightarrow C_1 C_2 C_3 C_4 \text{ par.}$$

$$(*) \quad K = (0, \beta)$$

$$K' = (0', \beta')$$

$$\beta = (i, j)$$

$$\beta' = (i', j')$$

$$[0']_K = \begin{bmatrix} 6 \\ 0 \end{bmatrix} \Rightarrow 00' = 6\vec{i} + 0\vec{j}$$

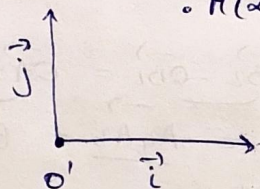
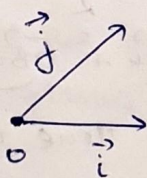
$$[i']_{\beta} = \begin{bmatrix} -3 \\ 1 \end{bmatrix} \Rightarrow i' = -3\vec{i} + 1\vec{j}$$

$$[j']_{\beta} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow j' = 1\vec{i} + 1\vec{j}$$

$$\bullet [A]_K = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, [B]_K = \begin{bmatrix} 6 \\ 4 \end{bmatrix}, [C]_K = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \rightarrow K'$$

• Base change matrix from $K \rightarrow K'$.

• $H(\alpha, \beta)$



$$\overrightarrow{OH} = \alpha \vec{i} + \beta \vec{j} \quad (1) \Rightarrow [H]_K = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$[H]_{K'} = ? \rightarrow \overrightarrow{O'H} = \overrightarrow{O'O} + \overrightarrow{OH} = -\overrightarrow{OO'} + \overrightarrow{OH} \quad \begin{matrix} (1) \\ (2) \end{matrix}$$

$$= -(6\vec{i} + 0\vec{j}) + \alpha\vec{i} + \beta\vec{j}$$

$$= (\alpha - 6)\vec{i} + \beta\vec{j}$$

$$= \begin{bmatrix} \alpha - 6 & \beta \end{bmatrix} \cdot \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix} \quad (*)$$

$$\begin{bmatrix} \vec{i} \\ \vec{j} \end{bmatrix} = \underbrace{\begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix}}_A \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix} \Rightarrow \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix} = A^{-1} \cdot \begin{bmatrix} \vec{i} \\ \vec{j} \end{bmatrix}$$

$$\begin{aligned} (\Psi): \vec{OH} &= [\alpha - 6 \quad \beta] \cdot A^{-1} \cdot \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix} = [\alpha - 6 \quad \beta] \cdot \begin{bmatrix} -1 & 1 \\ 1 & 3 \end{bmatrix} \cdot \frac{1}{4} \cdot \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix} = \\ &= \frac{1}{4} \cdot [\alpha - 6 + \beta \quad \alpha - 6 + 3\beta] \cdot \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix} \\ &= \frac{1}{4} \cdot [(6 - \alpha + \beta) \vec{i}' + (\alpha - 6 + 3\beta) \vec{j}'] \end{aligned}$$

↳ formula for changing from one frame to another.

$$A = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix}, \quad A^{-1} = \frac{1}{\det A} \cdot A^* = -\frac{1}{4} \cdot \begin{bmatrix} 1 & -1 \\ -1 & -3 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -1 & 1 \\ 1 & 3 \end{bmatrix}.$$

$$\det A = -4$$

$[H_{K \leftarrow K'}] = A^{-1} \rightarrow$ the matrix of change of basis from $K \rightarrow K'$. - drawing -

$$M_{K \leftarrow K'} = (M_{K' \leftarrow K})^{-1}.$$

$$[H]_K = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow [H]_{K'} = \begin{bmatrix} \frac{6 - \alpha + \beta}{4} \\ \frac{\alpha - 6 + 3\beta}{4} \end{bmatrix}$$

$$[A]_{K'} = \begin{bmatrix} \frac{6 - 0 + 2}{4} \\ \frac{0 - 6 + 3 \cdot 2}{4} \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$[A]_K = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$[B]_{K'} = \begin{bmatrix} \frac{6 - 6 + 4}{4} \\ \frac{6 - 6 + 3 \cdot 4}{4} \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$[B]_K = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$$[C]_{K'} = \begin{bmatrix} \frac{6 - 3 + 1}{4} \\ \frac{3 - 6 + 3}{4} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$[C]_K = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

*)

ABCD tetrahedron.

$$K_A = (A, \mathcal{B}), \quad \mathcal{B} = (\vec{AB}, \vec{AC}, \vec{AD})$$

$$K_B = (B, \mathcal{B}), \quad \mathcal{B} = (\vec{BA}, \vec{BC}, \vec{BD})$$

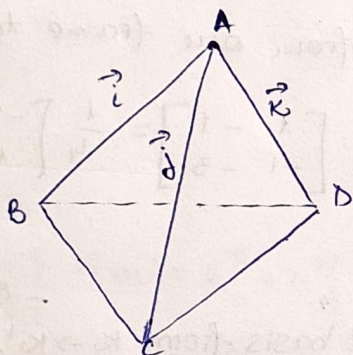
$$K'_A = (A, \mathcal{B}), \quad \mathcal{B} = (\vec{BA}, \vec{BD}, \vec{BC})$$

a) A, B, C, D w.r.t. K_A, K_B, K_C .

* b) base change matrix $K_{A'}$ to K_A and K_A to K_B .

* c) relative orientation of frames $\rightarrow \det \begin{cases} > 0 \Rightarrow \text{same or.} \\ < 0 \Rightarrow \text{opposite.} \end{cases}$

a) K_A :



$$\vec{AA} = 0 \cdot \vec{i} + 0 \cdot \vec{j} + 0 \cdot \vec{k} \Rightarrow A(0, 0, 0).$$

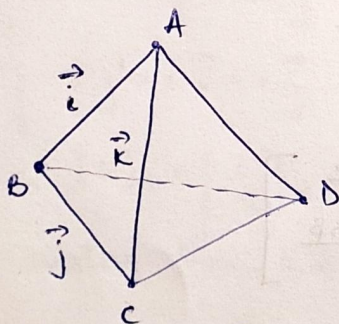
$$\vec{AB} = 1 \cdot \vec{i} + 0 \cdot \vec{j} + 0 \cdot \vec{k} \Rightarrow B(1, 0, 0)$$

$$C(0, 1, 0)$$

$$D(0, 0, 1)$$

(with respect to the first frame)

K_B :



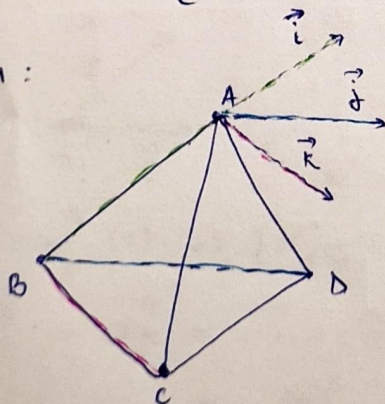
$$B(0, 0, 0)$$

$$A(1, 0, 0)$$

$$C(0, 1, 0)$$

$$D(0, 0, 1)$$

K'_A :



$$\vec{AB} = -\vec{i} \Rightarrow B(-1, 0, 0)$$

$$A(0, 0, 0)$$

$$\vec{AC} = \vec{AB} + \vec{BC} = -\vec{i} + \vec{k}$$

$$\Rightarrow C(-1, 0, 1)$$

$$\vec{AD} = \vec{AB} + \vec{BD} = -\vec{i} + \vec{j}$$

$$\Rightarrow D(-1, 1, 0).$$

b)

Obs! $\mathbb{E}^m \rightarrow m$ -dimensional Euclidean space.

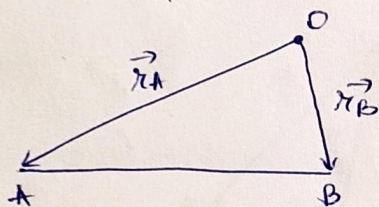
(usually $m=3$ or $m=2$)

$V^m \rightarrow$ space of vectors in $\mathbb{E}^m$

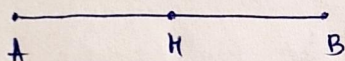
? ($\mathbb{R}$ vector space)

$$A_1, \dots, A_k \in \mathbb{E}^m \Rightarrow \overrightarrow{A_1 A_2} + \overrightarrow{A_2 A_3} + \dots + \overrightarrow{A_{m-1} A_m} = \vec{0}$$

If we fix $O \in \mathbb{E}^m \Rightarrow (\forall) A \in \mathbb{E}^m$ we can define $\vec{r}_A = \overrightarrow{OA} =$
= the position vector of A (w respect to O).



$$\vec{AB} = \vec{r}_B - \vec{r}_A$$



$A, H, B \in \mathbb{E}^m$; H midpoint of $[AB] \Leftrightarrow$

$$\Leftrightarrow \vec{r}_H = \frac{1}{2} (\vec{r}_A + \vec{r}_B)$$